

Functional disorders – diagnosis, causes and management: insights from brain imaging

Dr Quinton Deeley

Consultant Developmental Neuropsychiatrist

South London and Maudsley NHS Foundation Trust

Senior Lecturer

Institute of Psychiatry, Psychology and Neuroscience

Kings College London

What is a ‘functional’ symptom?

- ‘a problem due to a change in function (of the nervous system) rather than structure’ (Stone 2009)
- Non-epileptic / dissociative seizures
- Weakness
- Sensory symptoms
- Movement disorders
 - Tremor
 - Dystonia
 - Gait disorders
- Pain
- Fatigue
- Cognitive symptoms
- Visual symptoms
- Globus
- Dysphonia
- Drop attacks
- Back pain
- Fibromyalgia
- palpitations

A BRIEF DIS-
COURSE OF A DIS-
EASE CALLED THE
Suffocation of the
Mother.

Written vpon occasion which
hath beene of late taken thereby, to suspect pos-
session of an euill spirit, or some such like
supernaturall power,

*Wherin is declared that diuers strange
actions and passions of the body of man, which in
the common opinion, are imputed to the Diuell,
haue their true naturall causes, and do
accompanie this disease.*

By EDWARD IORDEN
Doctor in Physicke.



LONDON.

Printed by Iohn Windet, dwelling at the Signe of
the Crosse Keyes at Pawles Wharfe. 1603.

Frequency of 'conversion' symptoms (Stone 2009)

- ▶ Dissociative/non-epileptic seizures account for about 20% of referrals to “first fit” clinics and to specialist epilepsy clinics.
- ▶ Up to 50% of admissions to hospital with “status epilepticus” are in fact dissociative seizures/“pseudostatus”.
- ▶ Functional weakness has an annual incidence of at least 5/100 000, similar to multiple sclerosis.
- ▶ Functional movement disorders account for 5–10% of patients seen in a movement disorders clinic.

Clues to malingering or factitious disorder

(Stone 2009)

The main clues to malingering or factitious disorder are:

- ▶ inconsistency *in the history* on different occasions (between doctors or between relative and patient)
- ▶ an admission from the patient that they have lied about other things in the past
- ▶ avoidance of tests
- ▶ a direct confession
- ▶ evidence of gross inconsistency from covert surveillance (for example a wheelchair patient who is seen playing tennis)
- ▶ simulation of symptoms that mimics disease very closely (for example a patient who has tonic then clonic movements in their seizure rather than just shaking).

Aetiology of FNS

Carson, Hallett, Stone (2016)

The etiology of functional symptoms (functional neurologic disorder: FND)

Precipitating

FND is a disorder of sensorimotor processing in which erroneous health beliefs or expectations distort an, often noxious, somatosensory experience. This process is facilitated by misdirected and overly precise attention, anxiety, and dissociation. The symptom formation helps “make sense” of the amorphous somatic experience. The patient can be either consciously or preconsciously complicit in it

Perpetuating

Once present, FND can be perpetuated by maladaptive behavioral responses, both operant and classic learning, mood disorder, and central nervous system plasticity

Predisposing

Patients who have pre-existent mood/anxiety problems, excessive threat vigilance, or certain obsessive or rigid cognitive styles are more vulnerable; some of these risks may relate to the experience of abusive or aversive events currently, the recent past, or childhood. There is also a mild genetic risk and almost certainly other risk factors as yet unknown

The common sense model of illness regulation

Carson, Hallett, Stone (2016)

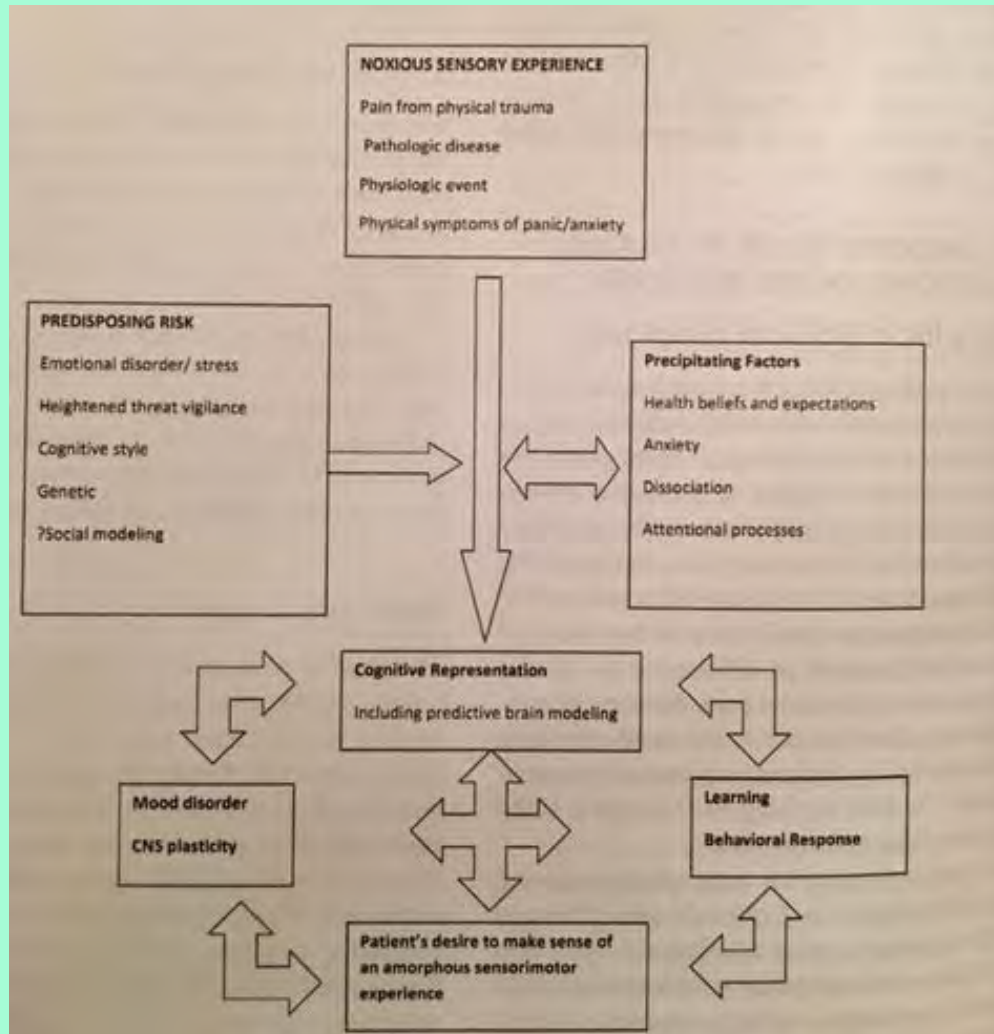
The common-sense model of illness regulation

Element	Cognition	Distortions	Example in functional disorders
Identity	What are these symptoms?	Symptoms cause labels But labels also lead to the self-generation of symptoms	<i>I have limb weakness; I think it is a stroke</i>
Cause	What caused these symptoms?	Beliefs that symptoms are due to damage and therefore irreversible	<i>A stroke is a clot in the brain; I think it happened because I was overdoing it</i>
Consequences	What effects will the symptoms have on my life?	Cog representations guide subsequent behavior	<i>I am scared I could end up in a wheelchair or even die</i>
Time line	How long will the symptoms last?	Behaving and adjusting life in the belief something will go on for ever, i.e., quitting job, can become a self-fulfilling prophecy	<i>I don't think my leg will ever get better. What if I get more disabled?</i>
Cure and control	What will help make the symptoms better?	Change in symptoms provides feedback on coping strategies and may result in reappraisal of symptoms or adoption of maladaptive strategies, i.e., pain on activity leading to increased downtime	<i>I think I need to rest more to make sure this doesn't happen again. It is important I don't do anything to provoke it or cause it to recur</i>

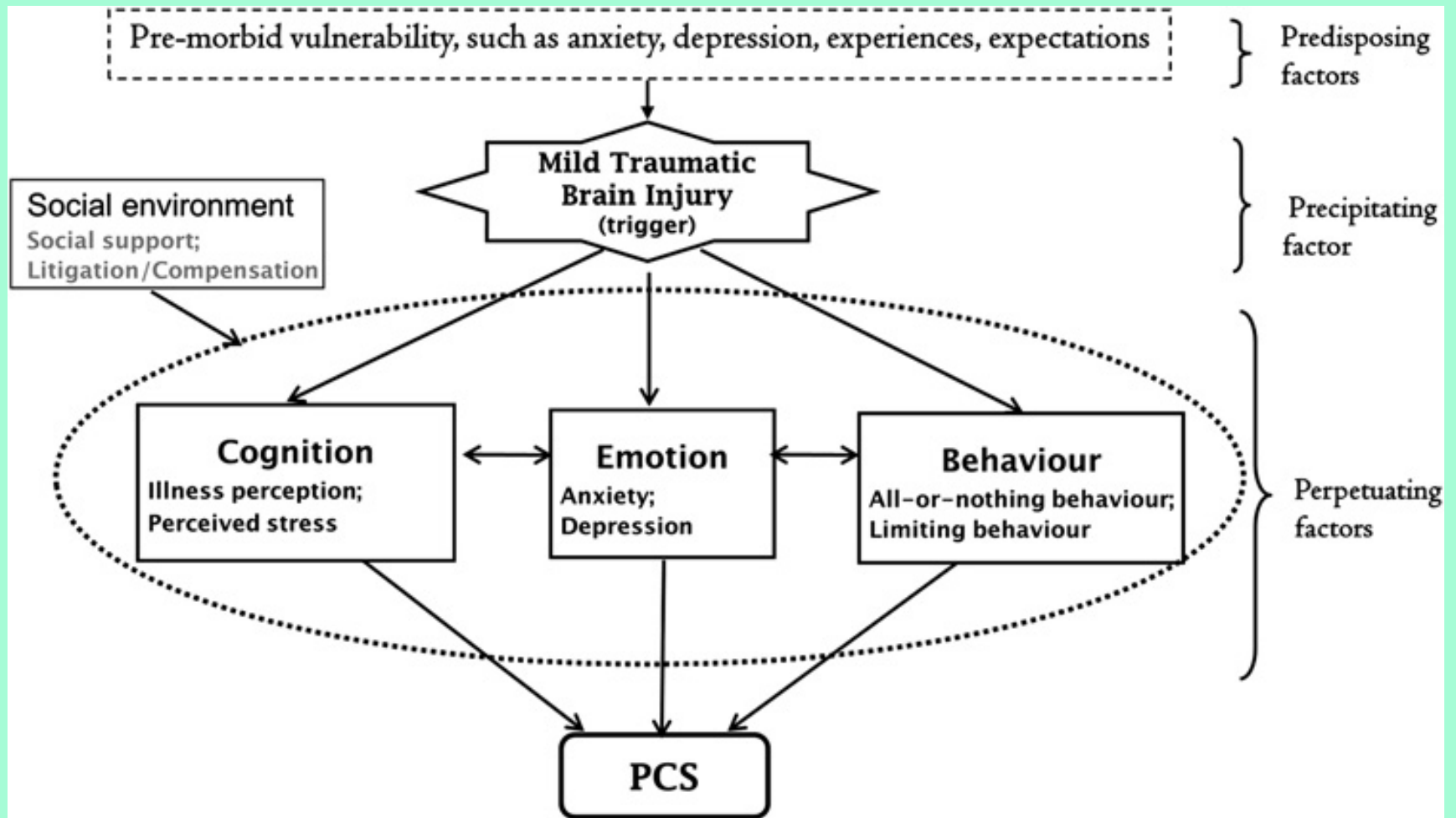
from Stone et al. (2009b), by courtesy of the Guarantors of Brain.

Causes of FNS

Carson, Hallett, Stone (2016)



The post-concussional symptom model (Hou et al 2011)



Mechanisms of functional neurological symptoms:

The power of belief

REMARKS

ON

PARALYSIS, AND OTHER DISORDERS OF MOTION AND SENSATION, DEPENDENT ON IDEA.*

By J. RUSSELL REYNOLDS, M.D., F.R.S.,
Professor of Medicine in University College; Physician to University College
Hospital; etc.

THE object of this paper is to show—1. That some of the most serious disorders of the nervous system, such as paralysis, spasm, pain, and otherwise altered sensations, may depend upon a morbid condition of emotion, of idea and emotion, or of idea alone; 2. That such symptoms often exist for a long time, appearing as complicated diseases of the brain or spinal cord; 3. That they resist many different kinds of treatment, being alike unmoved by sedatives and irritants, by attention or neglect, but that they disappear entirely upon the removal of the erroneous idea; 4. That they occur independently of anything that could be called either insanity of mind, hysteria, hypochondriasis, or malingering; 5. That they are often, but not constantly, associated with some bodily weakness or general debility; 6. That they sometimes associate themselves with distinct and definite diseases of the nervous centres, so that it becomes very important to know how much of a given case is due to organic lesion, and how much to morbid ideation; 7. That it is possible to make a diagnosis with regard to them in many instances; and, 8. That the principles upon which their treatment should be conducted are simple, and their application marvellously successful.

Remarks on paralysis and other disorders of motion and sensation dependent on an idea

J. Russell Reynolds 1869

“some of the most serious disorders of the nervous system, such as paralysis, spasm, pain, and otherwise altered sensations, may depend upon a morbid condition of emotion, or idea and emotion, or of idea alone ... they sometimes associate themselves with distinct and definite diseases of the nervous centers, so that it becomes very important to know how much a given case is due to an organic lesion, and how much to morbid ideation”

Jean Martin Charcot (d. 1893)

(painting by Andre Brouillet 1887)



Charcot's hypothesis

- Charcot - symptoms could derive from unconscious 'fixed' ideas based on suggestions or autosuggestions "remaining isolated from the rest of the mind and expressing themselves outwardly through corresponding motor phenomena" (Charcot)
- hysterical paralysis arose when "the idea comes to the patient's mind that he might become paralysed; in one word through autosuggestion, the rudimentary paralysis becomes real" (Charcot & Marie 1892)
- similar brain processes were responsible for the unexplained neurological symptoms of hysteria and the pseudo-neurological behaviours commonly produced by hypnosis.

Pierre Janet (1859 – 1947)



Janet and Dissociation (1907)

- a ‘contraction of the field of consciousness’ resulting in an abnormal compartmentalisation of mental functions that are normally closely associated
- Influence of the suggestive effect of ‘fixed ideas’ typically based on unresolved traumatic memories.
- not generally accessible to consciousness, but ‘emancipated’ in hysterical individuals who had an abnormal weakness of will and consciousness.
- ‘Ideas’ are ‘systems of images’ relating to movement, viscera, of other aspects of functioning.
- Hysterical individuals suggestible, contributing to symptom formation but also rendering them amenable to therapeutic suggestion

Hypnosis

(Heap et al 2000)

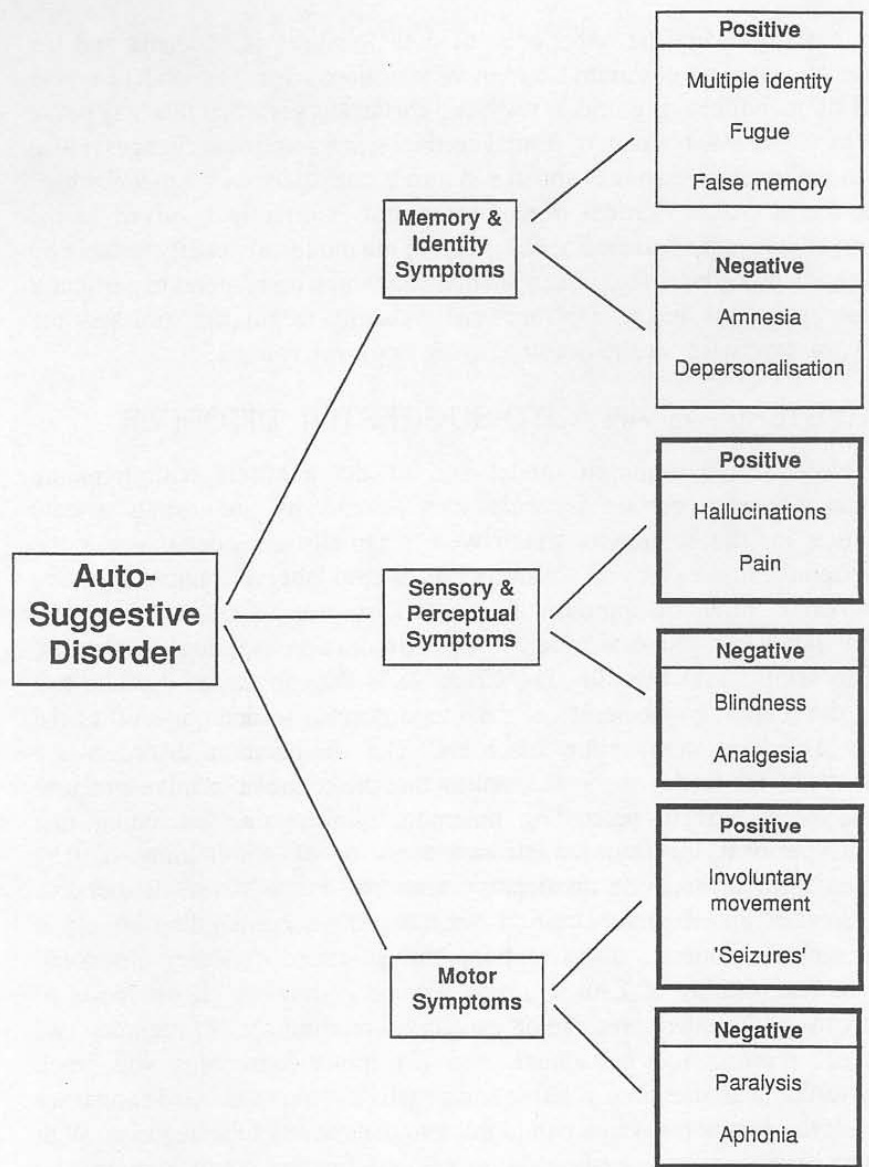
- controlled modulation of components of cognition
 - e.g. awareness, volition, perception, and belief
- by an external agent (the hypnotist), or oneself (self-hypnosis)
- using attentional focusing on ideas and images that may evoke the intended effects
- verbal communications used to achieve these effects = ‘suggestions’
- Classic suggestion effect – responses experienced as involuntary, effortless

Types of Suggestion

- Ideomotor
 - Involuntary limb movement
- Challenge
 - Subject unable to perform a movement, even when challenged to do so (eg raise arm)
- Cognitive
 - Complex changes in experience
 - Hallucinations, amnesia, analgesia

Suggestion

- “a form or type of communicable belief capable of producing and modifying experiences, thoughts and actions. Suggestions can be (a) intentional/ non-intentional, (b) verbal/nonverbal, or (c) hypnotic/ non-hypnotic” (Halligan and Oakley 2014).
- Intentionally administered verbal suggestions can produce suggested effects outside a hypnotic context - ‘imaginative suggestibility’ (Braffmann and Kirsch 1999).



Brain correlates of functional
symptoms, and the role of symptom
modelling with suggestion

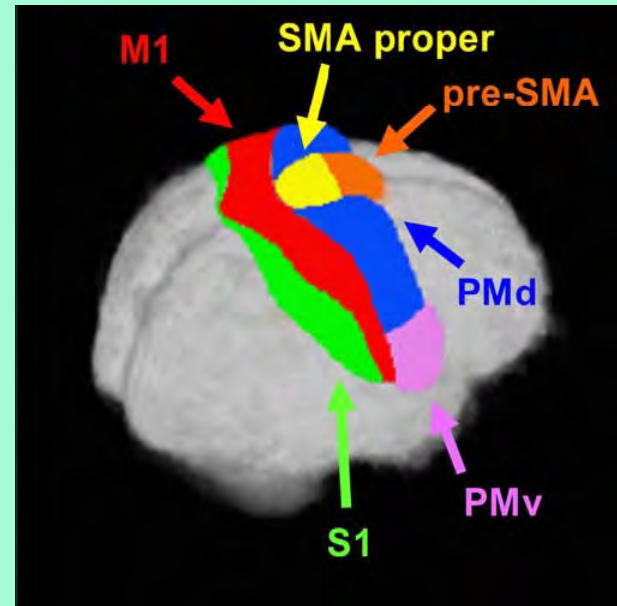
Dissociative and Suggested Limb Paralysis

- Similarities
 - Both conversion disorder patients and people with hypnosis report subjectively real experience of loss of normal function
 - No evidence of underlying neurological abnormality
 - Symptoms conform to expectations and beliefs
- Differences
 - Temporal duration
 - Suggestions administered by hypnotist; no evidence of comparable process in conversion patients
 - A form of autosuggestion? Unconscious?

Dissociative and Suggested Limb Paralysis

- **PET studies**
- Compared to normal limb, attempted movement in
 - Paralysed limb with conversion disorder (Marshall et al 1997)
 - Paralysed limb due to suggestion (Halligan et al 2000)
- increased activation
 - contralateral anterior cingulate
 - Contralateral orbitofrontal cortex
- decreased activation
 - primary motor cortex
- **fMRI**
- Cojan et al. - involvement of brain regions involved in affective modulation of motor function (OFC, vmPFC, via VLPFC), and processing external and internal stimuli relevant to the self (vmPFC and precuneus).

Regions More Active During Attempted Movement of Paralysed Limb compared to Limb Movement in Hypnotic State (Deeley et al 2013)



Mayka et al (2006)

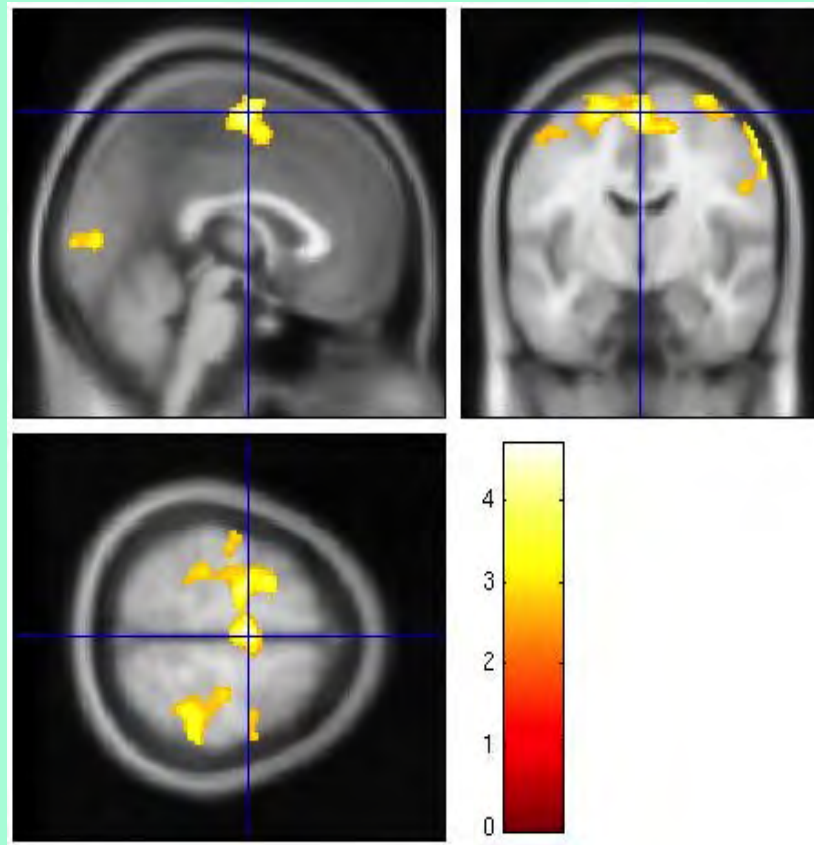
Other mechanisms for motor paralysis symptoms? (Vuilleumier 2014)



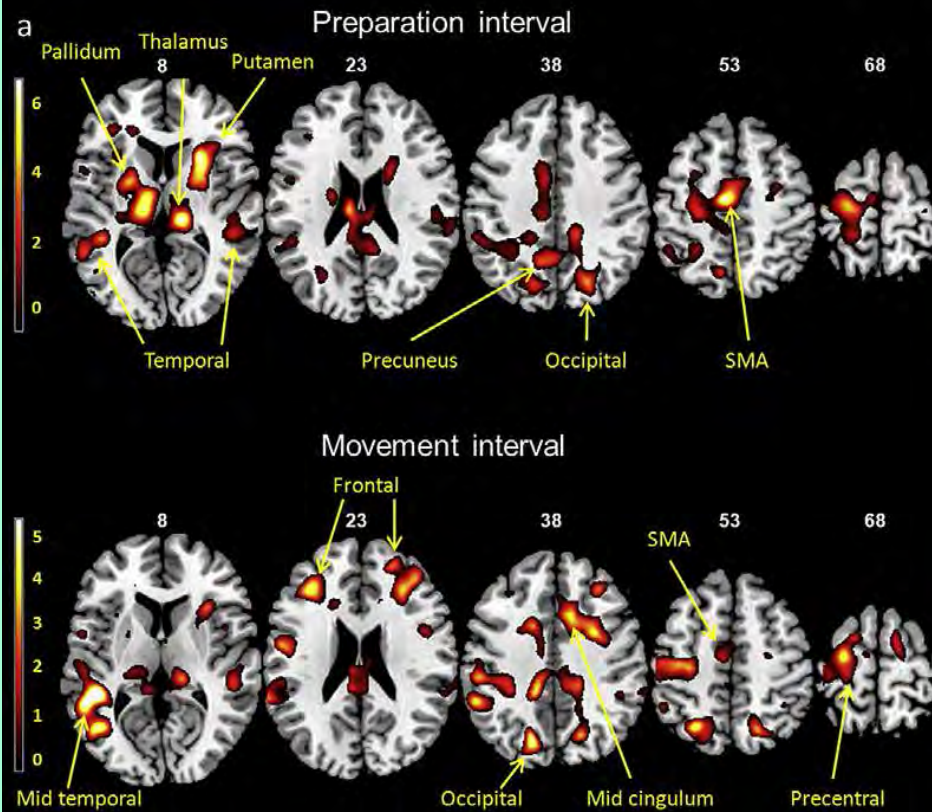
Figure 6 Schematic neuroanatomical model of psychogenic paralysis. Both precuneus and ventromedial prefrontal cortex (VMPFC) are core components of brain networks mediating access to self-representations and personal episodic memories, including sensory or agency information (for precuneus) and affective relevance (for VMPFC). Frequent recruitment of these areas in conversion patients and enhanced connectivity with motor or sensory areas suggests an important role in modulating behavior and self-awareness, perhaps by modulating motor or sensory processes (at cortical and/or subcortical level) under the influence of memory and affective representations unconsciously generated in these regions. VMPFC activity appears specific to conversion, whereas precuneus is also recruited during hypnosis.

Motor areas showing reduced functional connectivity with SMA during involuntary compared to voluntary movement.

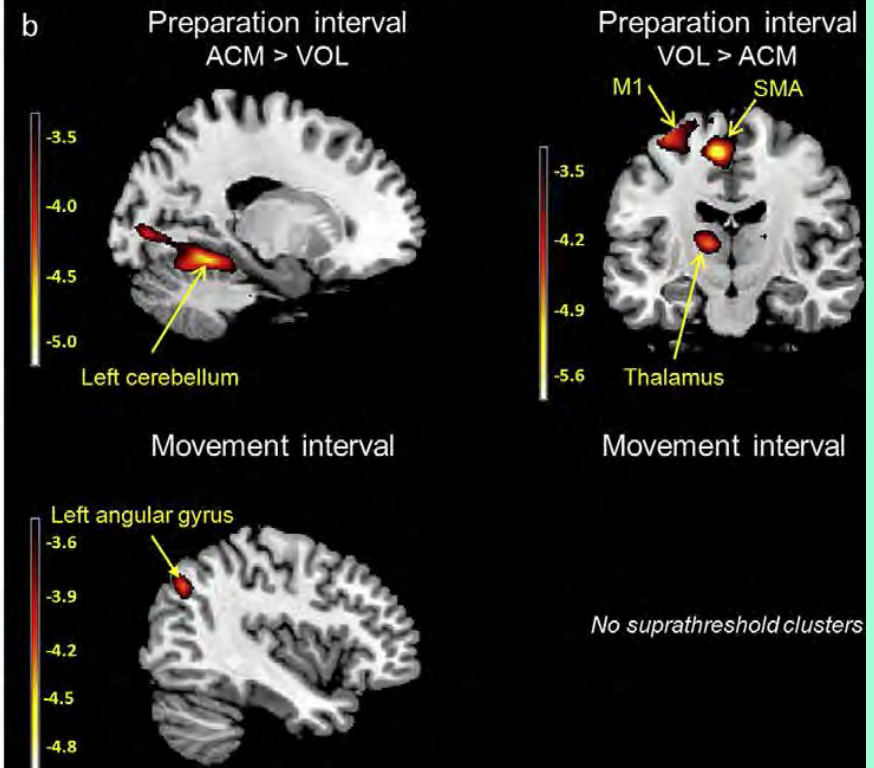
Crosshairs show peak activation in (left) SMA.



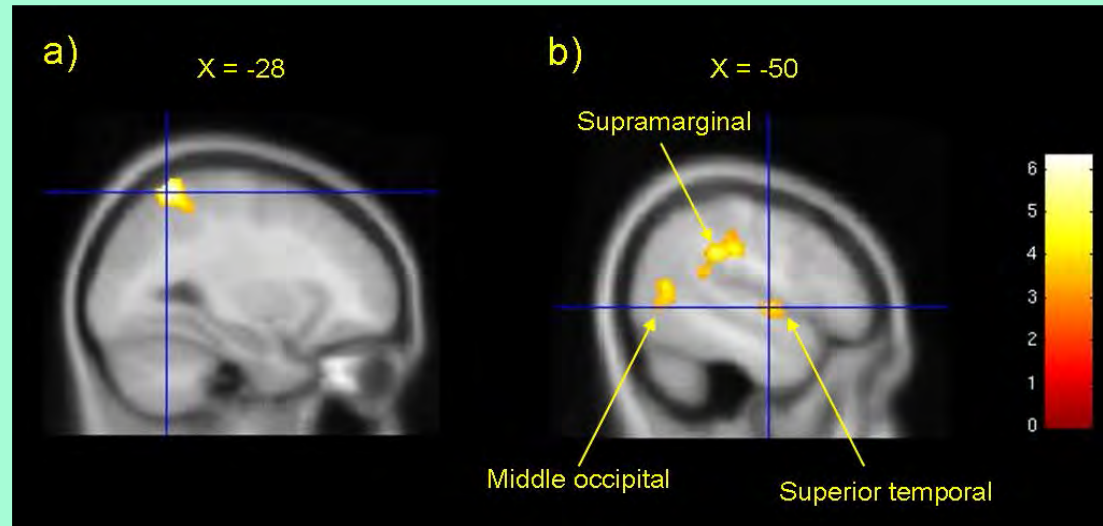
Thought insertion



Alien control of movement



Involuntary movement, regions more active when aware compared to unaware



a) left SPL (BA 7) and IPL (BA 40)

Functional symptoms involve different brain processes depending on the type of symptom (Aybek and Vuilleumier 2016, Deeley 2016)

- Imaging studies of patients, and symptom modeling studies with suggestion and fMRI, present evidence of several processes depending on the symptom type, and study design
- Limb paralysis – failure to generate normal motor intention, or involuntary inhibition of attempted movement?
- Involuntary movement – reduced activity in motor control regions (SMA), and altered sensory processing of movement
- Thought insertion/ involuntary thoughts – reduced activity in SMA
- Memory loss – reduced perfusion in memory systems at rest, but also frontal inhibition of memory systems during attempted memory retrieval
- Loss of awareness of limb movement – loss of sensory processing
- Pain – activation of pain matrix in the absence of pain stimuli
- Dystonia – abnormal movement generation (increased basal ganglia activity)

Are functional and suggested symptoms due to feigning?

- All studies that have compared functional and suggested symptoms to feign or imagine symptoms have shown different activation patterns (Aybek and Vuilleumier 2016; Deeley 2016)
 - Limb paralysis (functional and suggested), tremor (functional), pain (suggested), and involuntary limb movement (suggested)
- Functional and suggested symptoms are subjectively real, distinct from ‘simulation’, ‘feigning’, and ‘imagining’

Integrative models of suggestions

- fMRI –
 - immediate changes in brain activity underpinning suggested effects
 - The end point of a broad set of changes in information processing
- suggestions alter experience by
 - mobilising representations from memory systems and altering causal attributions during pre-conscious processing
 - this alters the content of what is provided to our highly edited subjective version of the world.
 - consciousness and behaviour can be ‘distorted by disproportionately active material in memory, leaving us prone to both misperceptions and behaviors that conflict with goals in self-awareness’ (Brown et al 2011)
 - Predictive modelling / Bayesian inference – abnormal predictions alter sensory processing (Edwards 2016)

Talking about FNDs with patients (Goldstein and Mellers 2016)

Conveying a diagnosis of functional neurologic disorder

Explanation

Describe what the patient doesn't have (e.g., epilepsy, stroke)
and the reasons for drawing this conclusion

Describe what the patient does have – give the condition a name
and a description

Reassurance

Tell the patient that this condition is common

Tell the patient that s/he is not being accused of "putting on" the
symptoms

Mechanism

Describe dissociation

Describe the role of attention

Cause

Often not clear, complex

Summarize predisposing/triggering/maintaining factors that
have emerged in history taking and provide a description of
how these might be relevant and related to symptoms

Treatment

The patient's symptoms are potentially fully reversible

Direct the patient to self-help websites

Initiate medication withdrawal where appropriate (e.g.,
antiepileptic drugs)

Describe distraction techniques

Describe treatment

Cognitive Behavioural Therapy

Table 4 Examples of changes in thoughts and behaviour that can help in patients with functional symptoms

	Dissociative seizures	Functional weakness	Chronic back pain
Old thought	"Oh no—what's happening to me. Am I going to die during one of these attacks?"	"I've got MS; I'm going to end up in a wheelchair. No one believes me"	"My spine is damaged, I must avoid moving too much in case it makes it worse"
New thought	"I'm having something a bit like a panic attack"	"Mmm ... this is odd but it looks as if I can get better."	"My bones are fine, its my muscles that are stiff and out of condition"
Old behaviour	Avoid going out, tendency to succumb to blackout as a way of getting rid of horrible warning symptoms	Seeing lots of specialists, not doing very much in case it makes it worse	Avoiding exercise/back movement
New behaviour	Try out distraction techniques during warning symptoms	Gradually exercise, learn to expect relapses	Gradually exercise, change sleep patterns

Physiotherapy for FNDs

(Nielsen 2016)

Treatment principles commonly described in the literature

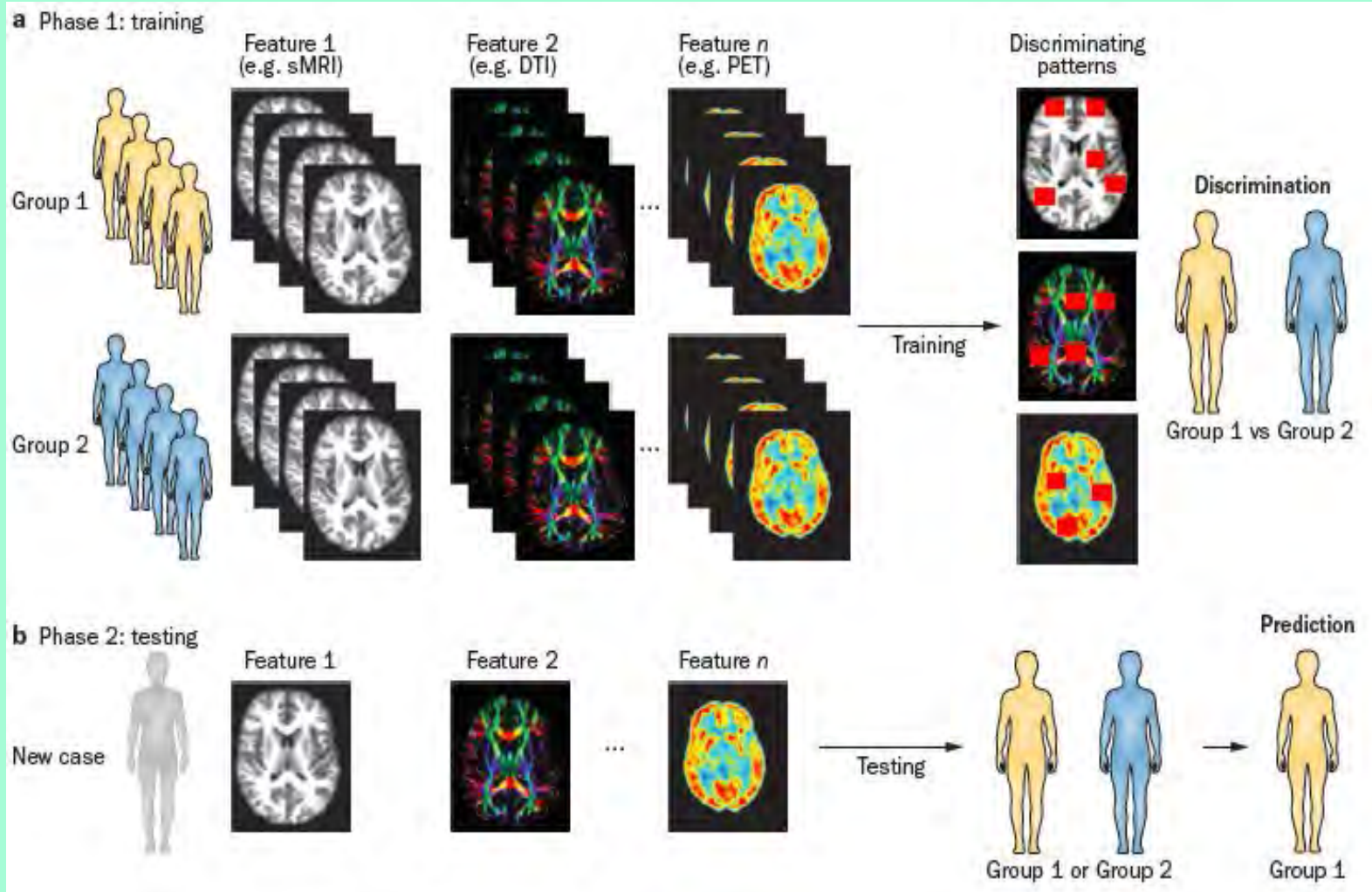
Build trust and rapport before challenging the patient
Create an expectation of recovery
Open and consistent communication
Involve family members
Avoid passive treatments
Encourage early weight bearing
Foster independence and self-management
Goal-directed rehabilitation
Avoid adaptive equipment
Use principles of behavioral management
Recognize and challenge unhelpful thoughts and behaviors
Develop a relapse prevention and management plan

Reproduced from Nielsen et al. (2015a), with permission from BMJ Publishing Group.

Diagnostic Imaging for FNDs?

Multivariate pattern classification

Ecker and Murphy 2014 NRNeuro

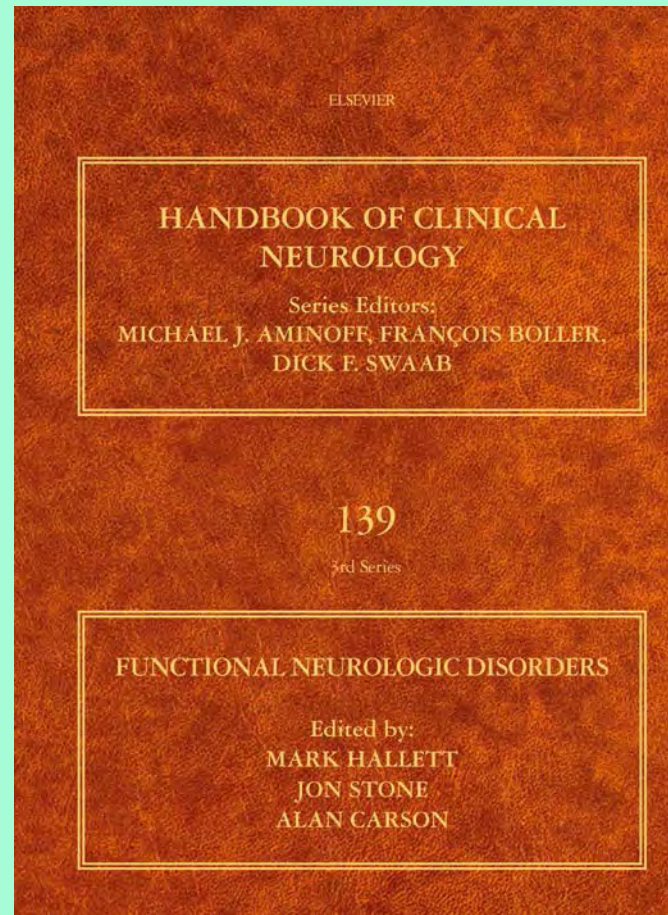


Conclusions

- Functional neurological symptoms are common and disabling
- Stigmatising and relatively neglected in medical training and research
- Subjectively real, not due to ‘feigning’ or ‘imagination’
- Beliefs and expectancies influence symptom formation through effects on brain function
- Brain correlates vary with the symptom
- Growing evidence base for CBT and physical treatments
- Diagnostic imaging – many challenges, but may help treatment stratification and prognosis more than diagnosis

Hallet, Stone, Carson (2016)

an up to date textbook



Research Team

- Dr Eamonn Walsh
- Dr Mitul Mehta
- Dr Vaughan Bell
- Luke Sheridan Raines
- Jasmit Jheeta
- Professor David Oakley, Hypnosis Unit, UCL
- Prof Peter Halligan, Cognitive Neuropsychology, University of Cardiff
- Dr Brian Toone, Maudsley Hospital/ IOP
- Centre for Neuroimaging Sciences, IOP
- Prof Steve Williams
- Vincent Giampietro
- Prof Mick Brammer

Discussion



Première hypnotisation de Braid.